CAPSM: Cognitive Adaptive Power System Management

A Brain-Inspired Dual-Process Architecture for Modern Grid Control

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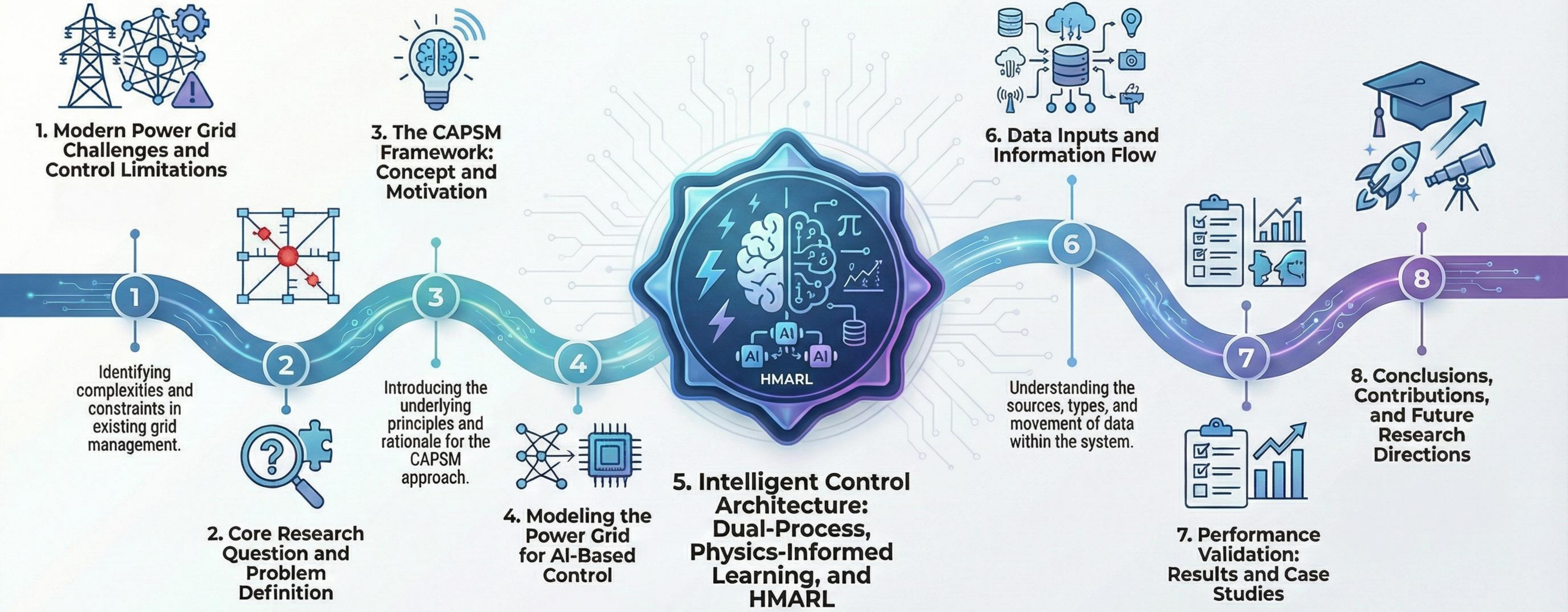
CAPSM: A Roadmap for Intelligent Power Grid Control



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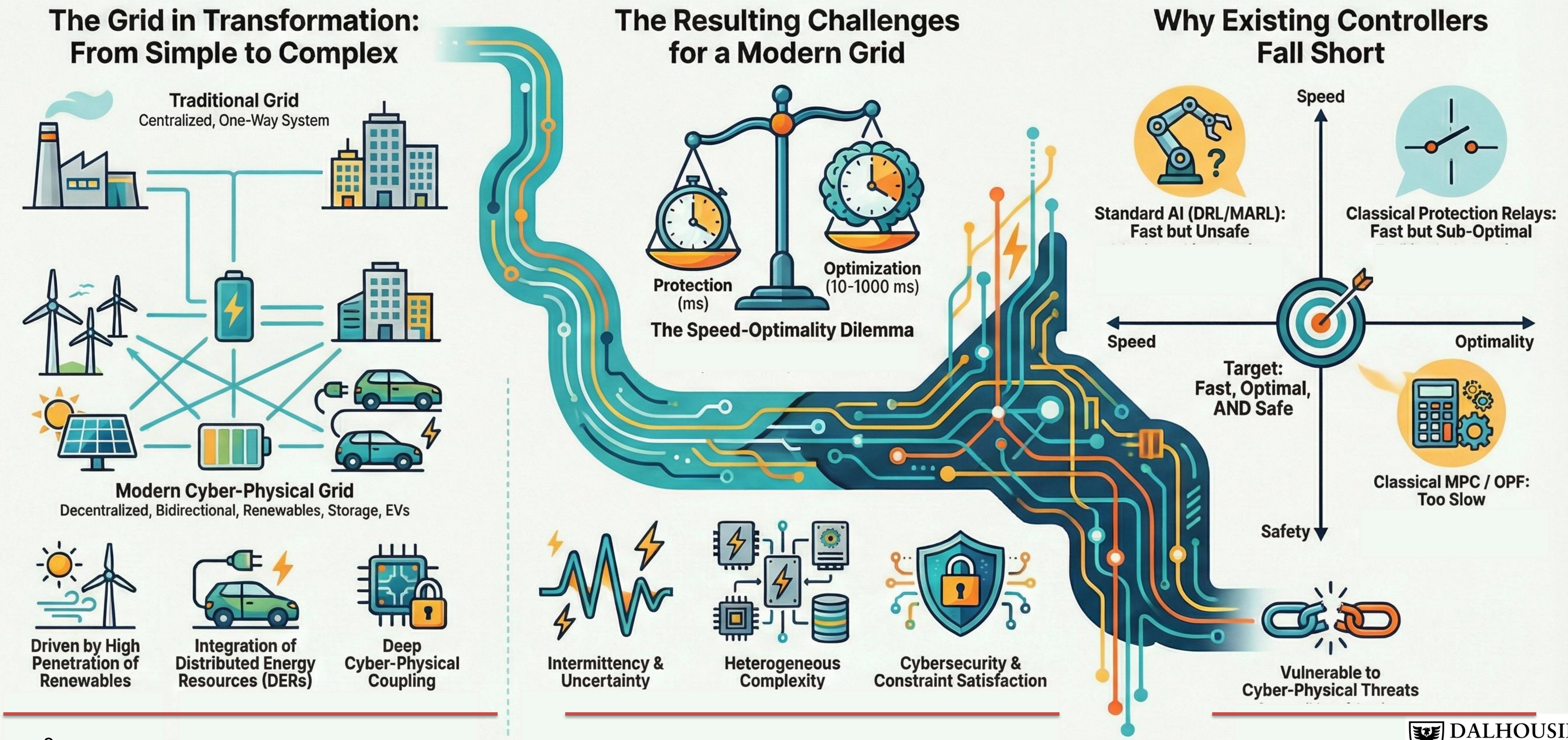
The Modern Power Grid's Dilemma: Fast, Smart, and Secure



This Research Addresses Five Key Gaps with a Unified Control Architecture

1. **Separation of Control**: The disconnect between fast protection and slower optimization.
2. **Lack of Physics in AI**: No inherent mechanism for embedding physical laws into real-time learning.
3. **Static Control Logic**: Absence of metacognitive adaptability in controllers.
4. **Poor Scalability**: Limited effectiveness of standard MARL in large, complex grids.
5. **Fragmented Security**: Ad-hoc treatment of cyber-physical security threats.

Problem Statement

How can we design a unified dual-process control architecture that provides millisecond-scale protection and multi-timescale optimal decision-making, while **embedding physics**, metacognition, and scalability in renewable-rich, cyber-physical grids?

Central Hypothesis

A dual-process, physics-informed, metacognitively guided architecture (**System-1 + System-2**) can simultaneously achieve the required responsiveness, safety, and scalability.

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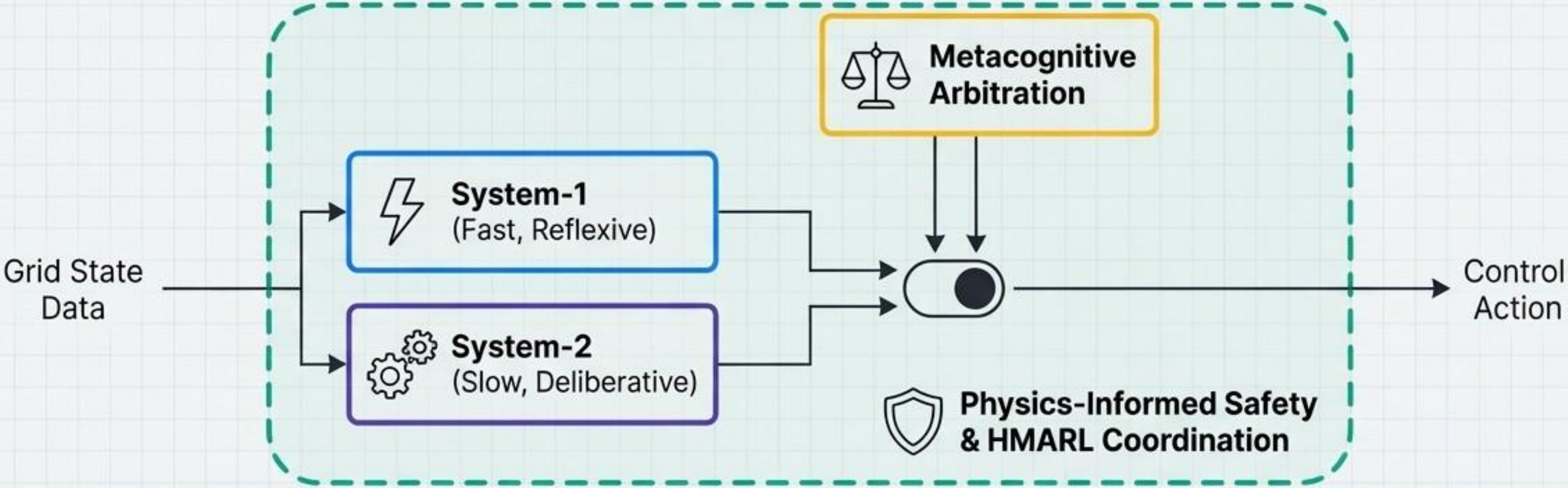
CAPSM Introduces a Brain-Inspired, Dual-Process Architecture with Metacognitive Arbitration



Governing Policies

System-1 Policy: $\pi_1(a|x) = P(a_t = a \mid x_t = x, \text{System}{-}1)$

System-2 Policy: $\pi_2(a|x) = P(a_t = a \mid x_t = x, \text{System}{-}2)$

Effective CAPSM Policy: $a_t \sim \pi_{\text{CAPSM}}(\cdot|x_t) = \pi_{\sigma_t}(\cdot|x_t)$, where $\sigma_t \in \{1, 2\}$

Component Summary

- **System-1:** CNN-LSTM based, provides ms-level, protection-like responses.
- **System-2:** QIRL-based, handles long-term optimization, planning, and coordination.
- **Metacognition:** Chooses which policy to trust based on confidence, novelty, and urgency.

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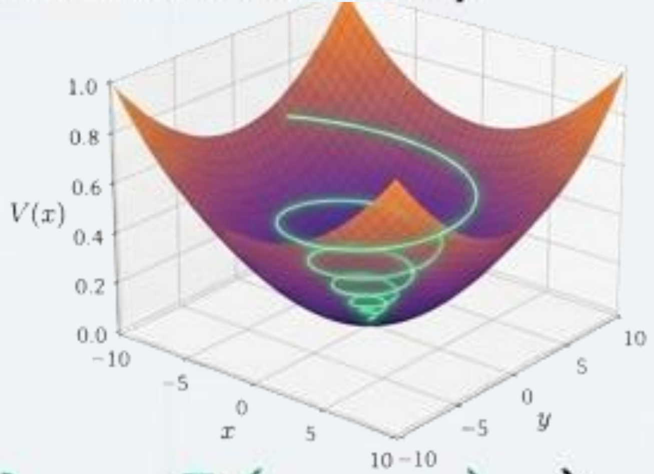
We Model the Grid as a Multi-Timescale MDP Governed by Stability Constraints

Core Formulation: Markov Decision Process (MDP)

$$\mathcal{M} = (S, A, P, r, \gamma)$$

Annotated State Evolution: $x_{t+1} = f(x_t, u_t, w_t)$

- $x_t \in S$: State (bus voltages, angles, frequency, SoC)
- $u_t \in A$: Action (FACTS setpoints, EV charging, generator controls)
- w_t: Disturbances (renewable fluctuations, faults)



The Objective: Reward Function

$$r(x_t, u_t) = -\Big(\underbrace{x_t^T Q x_t}_{\text{[Minimize State Deviations]}} + \underbrace{u_t^T R u_t}_{\text{[Minimize Control Effort]}} + \underbrace{\lambda_{\text{viol}}\, \Phi(x_t, u_t)}_{\text{[Penalize Constraint Violations]}} \Big)$$

Time-Scale Separation

Defines the dual operational speeds:

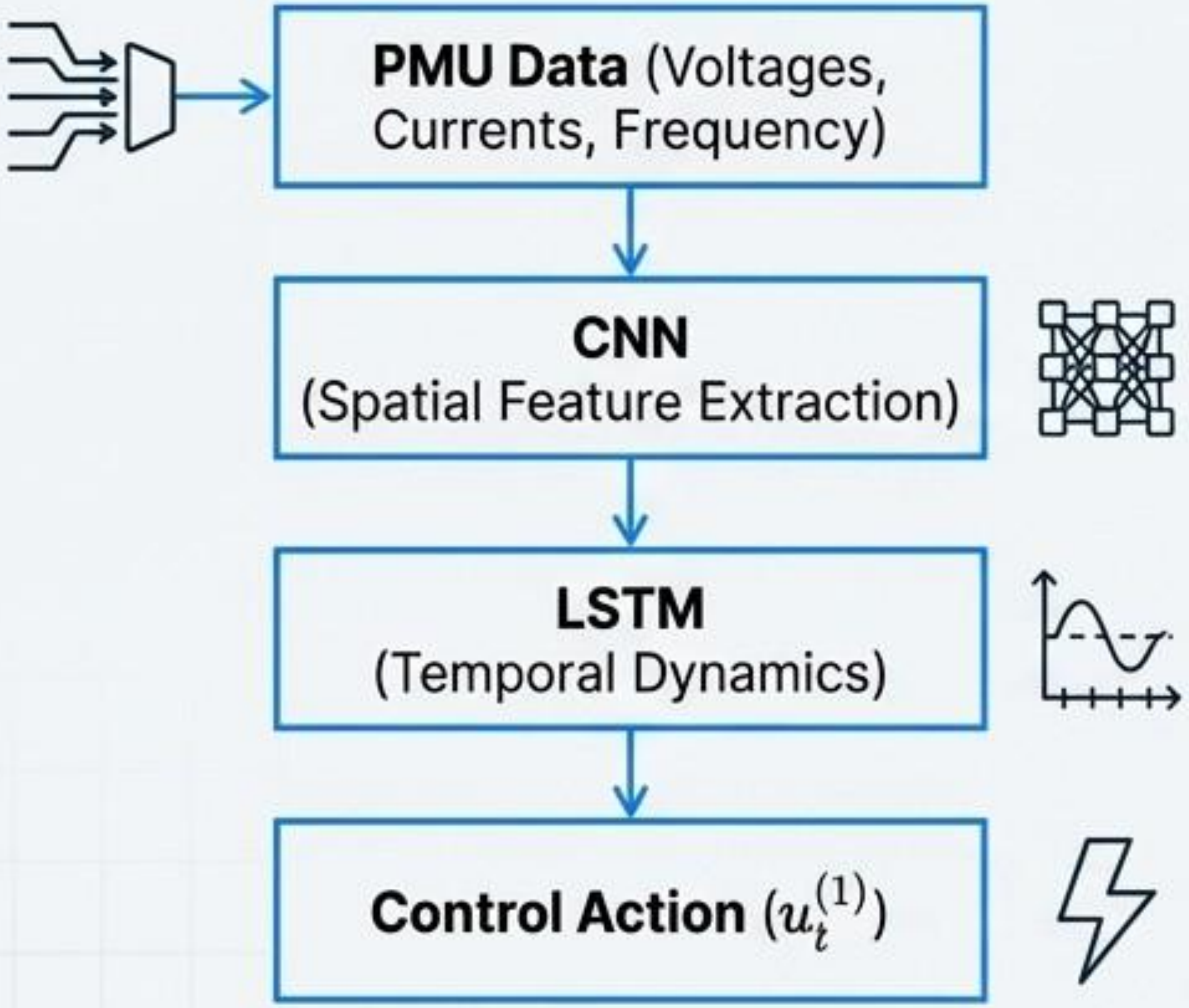
$$0 < \tau_1 \ll \tau_2, \text{ with } \tau_1 \approx O(\text{ms}),$$
$$\tau_2 \approx O(10\text{–}100\,\text{ms})$$

Lyapunov Stability Condition

The fundamental safety constraint the controller must respect:

$$V(x_{t+1}) - V(x_t) \leq -\alpha |x_t|^2,\ \alpha > 0$$

System-1 Provides Millisecond, Reflexive Control Using a CNN-LSTM Architecture



Architecture & Function

- **Inputs**: High-frequency PMU data streams (30–120 Hz) representing the grid's instantaneous state.
- **CNN Layer:** Extracts spatial correlations between different buses and grid locations.
- **LSTM Layer:** Captures temporal dynamics and identifies transient signatures of events.

Key Equations

Feature Extraction: $z_t = \mathrm{LSTM}(\mathrm{Conv}(X_{1:t}))$

Control Output: $u_t^{(1)} = g_\theta(z_t)$

Control Rationale

- System-1 primarily adjusts reactive power for voltage support, leveraging the near-instantaneous physical relationship: $\Delta V \approx (X/V)\Delta Q$.
- **Primary Role:** Fault detection, transient stabilization, and immediate voltage/frequency support.

System-2 Employs Quantum-Inspired Reinforcement Learning for Deliberative, Optimal Planning

Quantum-Inspired Reinforcement Learning (QIRL)

QIRL uses principles of quantum superposition and tunneling to achieve superior exploration and faster convergence in complex, non-convex operational landscapes compared to classical DRL.

Key Equations

Policy as Superposition:

$$|\psi_t\rangle = \sum_{a \in A} c_{t,a} \, |a\rangle, \text{ where } \pi(a|x_t) = |c_{t,a}|^2$$

Bellman Update:

$$Q_{k+1}(x_t, a_t) = (1-\beta)Q_k(x_t, a_t) + \beta[r_t + \gamma \max_{a'} Q_k(x_{t+1}, a')]$$

Amplitude Update with Tunneling:

$$c_{t+1,a} = c_{t,a} - \eta \frac{\partial L}{\partial c^*_{t,a}} + \kappa \, T(c_{t,\cdot}) \longleftarrow$$

Tunneling mechanism avoids local optima

Benefits

- Faster, more stable convergence (**~58%** fewer episodes)
- Better exploration of complex solution spaces

Used For

- Economic dispatch
- Vehicle-to-Grid (V2G) scheduling
- FACTS device coordination

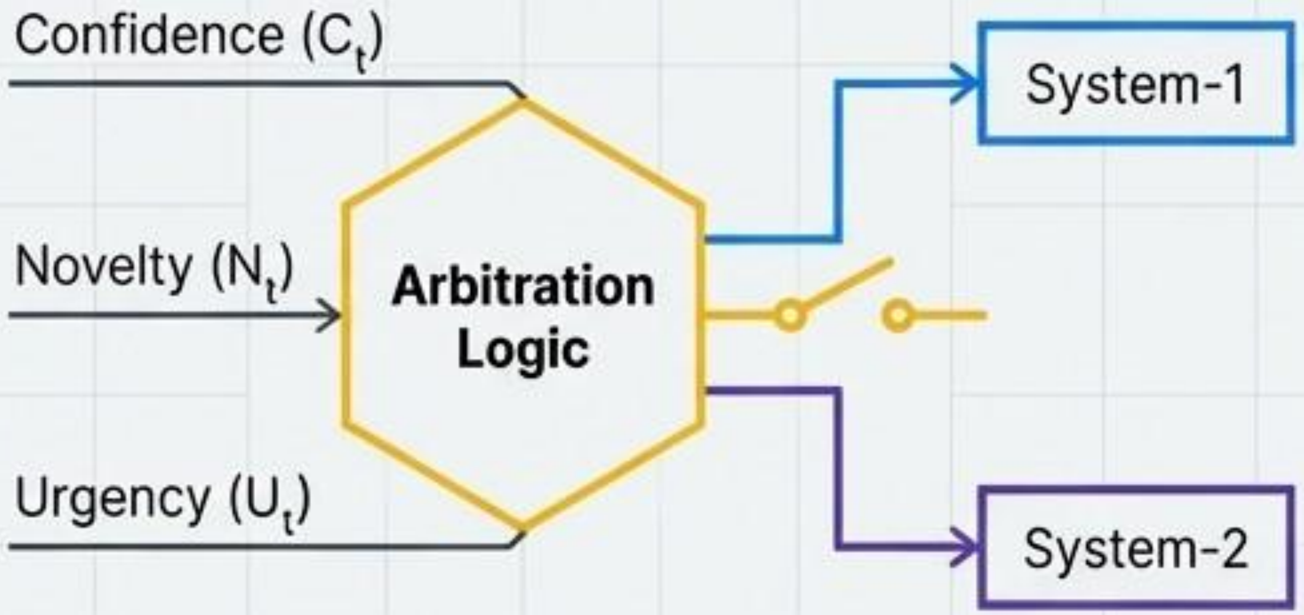
A Metacognitive Layer Intelligently Arbitrates Between Reflex and Deliberation

The Arbitration Rule

$$\sigma_t = \begin{cases} 1 & \text{(use System-1), if } C_t \geq \theta_C \text{ and } U_t \leq \theta_U \\ 2 & \text{(use System-2), if } N_t \geq \theta_N \text{ or } U_t > \theta_U \end{cases}$$

Defining the Triggers

- **Confidence (C_t):** $1 - \mathrm{Var}[Q(\mathbf{x}_t, \cdot)]$ (How sure is the model?)
- **Novelty (N_t):** $\|\mathbf{x}_t - \hat{\mathbf{x}}_{\text{train}}\|$ (Is the state unfamiliar?)
- **Urgency (U_t):** Severity of state deviation or event.



Narrative Explanation

- High confidence and low urgency allow the fast, efficient System-1 to operate.
- High novelty or urgency triggers the more robust, deliberative System-2 to analyze the situation and plan a new strategy.
- This is the mechanism that provides "cognition" and "self-awareness" to the control logic.

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Physics-Informed Neural Networks Are Embedded to Enforce Hard Safety Constraints

To enforce Kirchhoff's laws and physical operational limits (voltage, thermal, SoC) directly within the AI learning process, not as an afterthought.

The PINN Loss Function

$$L_{\mathrm{PINN}} = \lambda_{\mathrm{data}} \|y_{\mathrm{meas}} - y_\theta(\mathbf{x})\|^2$$ [Match Real-World Data]

$$+ \lambda_{\mathrm{PF}} \|f_{\mathrm{PF}}(\mathbf{x}, \mathbf{u})\|^2$$ [Obey Power Flow Physics]

$$+ \lambda_{\mathrm{op}} \Psi(\mathbf{x}, \mathbf{u})$$ [Respect Safety Limits]

Penalty Function Breakdown ($\Psi(\mathbf{x}, \mathbf{u})$)

$$\Psi(\mathbf{x}, \mathbf{u}) = \sum [\max(0, |V_i| - V_i^{\max})]^2$$ (Voltage Limits)

$$+ \sum [\max(0, |I_\ell| - I_\ell^{\max})]^2$$ (Thermal Limits)

$$+ \sum ([\max(0, \mathrm{SoC_j} - \mathrm{SoC}^{\max})]^2 + [\max(0, \mathrm{SoC}^{\min} - \mathrm{SoC_j})]^2)$$ (Storage Limits)

Key Takeaway

- PINNs are integrated into *both* System-1 and System-2.
- Target for constraint satisfaction is ≈99.8% compliance.

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Hierarchical Multi-Agent RL and Data Fusion Ensure Scalability and Robustness

Hierarchical Structure (HMARL)

Consensus Update: Local agents share information to converge on a coordinated state: $$x_i^{k+1} = x_i^k + \alpha \sum_{j \in N_i} w_{ij}(x_j^k - x_i^k)$$

Reward Structure: Local rewards (r_i) contribute to a global reward (r_{global}) for system-wide efficiency: $$r_{\text{global}} = \sum r_i$$

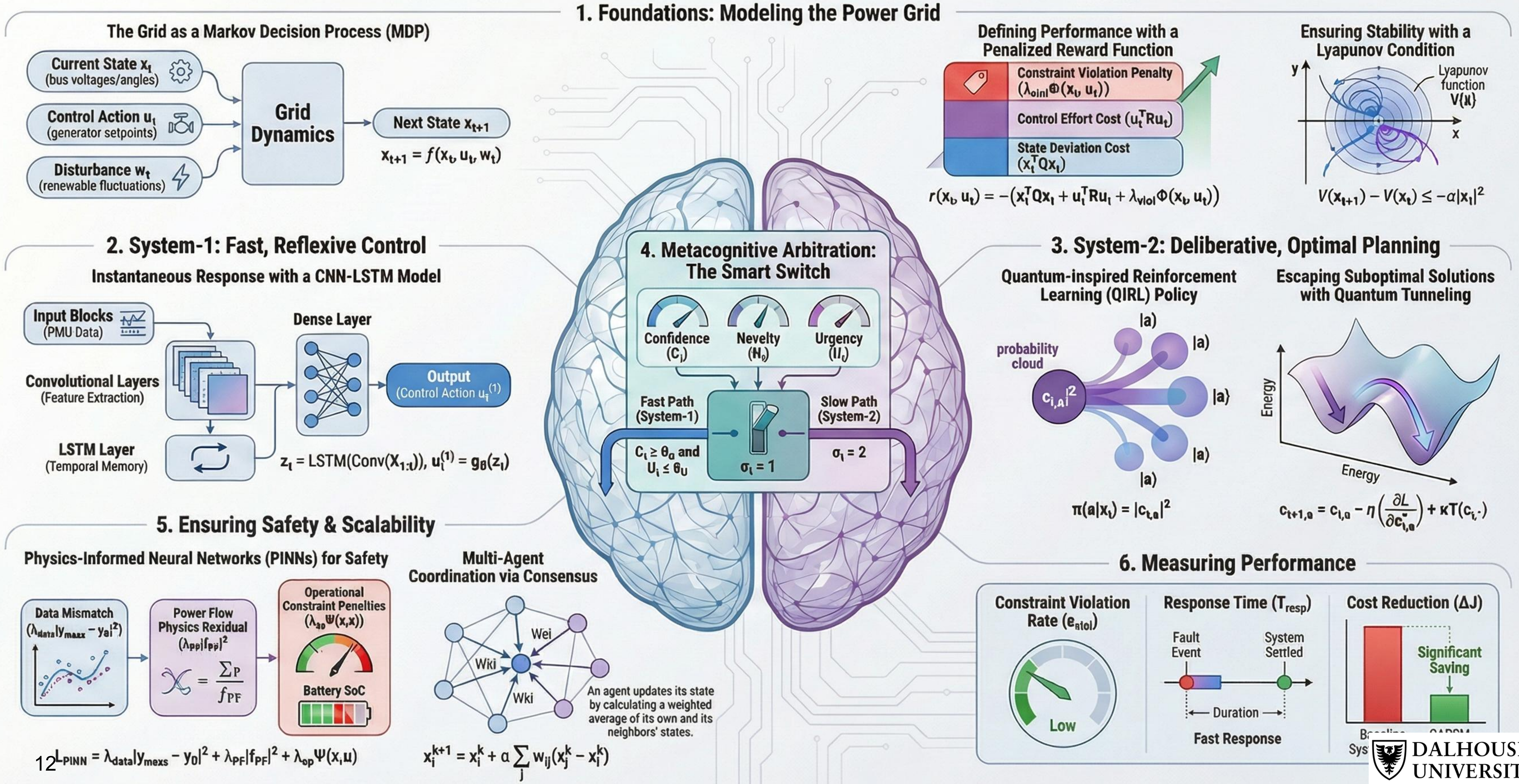
Robust Data Fusion

- **Challenge:** Fusing high-speed PMU data (30-120 Hz) with slower SCADA data (2-4 s) and forecasts.
- **Solution:** Uncertainty-aware weighted fusion. The weight w_k for each data source is inversely proportional to its variance σ_k^2, creating a unified state estimate.

Weight Calculation: $w_k = \dfrac{\sigma_k^{-2}}{\sum_j \sigma_j^{-2}}$

Fused State Estimate: $\hat{z}_{\text{fused}} = \sum_k w_k \hat{z}_k$

The Math Behind CAPSM: A Visual Guide to AI-Powered Grid Management



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Our Hypothesis Was Validated Through a Five-Phase Methodology on Standard IEEE Test Systems

Phase I: System-1 Prototyping & Fast Response (IEEE 9-bus)

Phase II: QIRL & System-2 Deliberation (9- and 39-bus)

Phase III: PINN Integration for Safety & Stability

Phase IV: HMARL Extensions for Scalability (39-, 118-bus)

Phase V: Integrated CAPSM Benchmarking (118-, 300-bus)

Test Systems

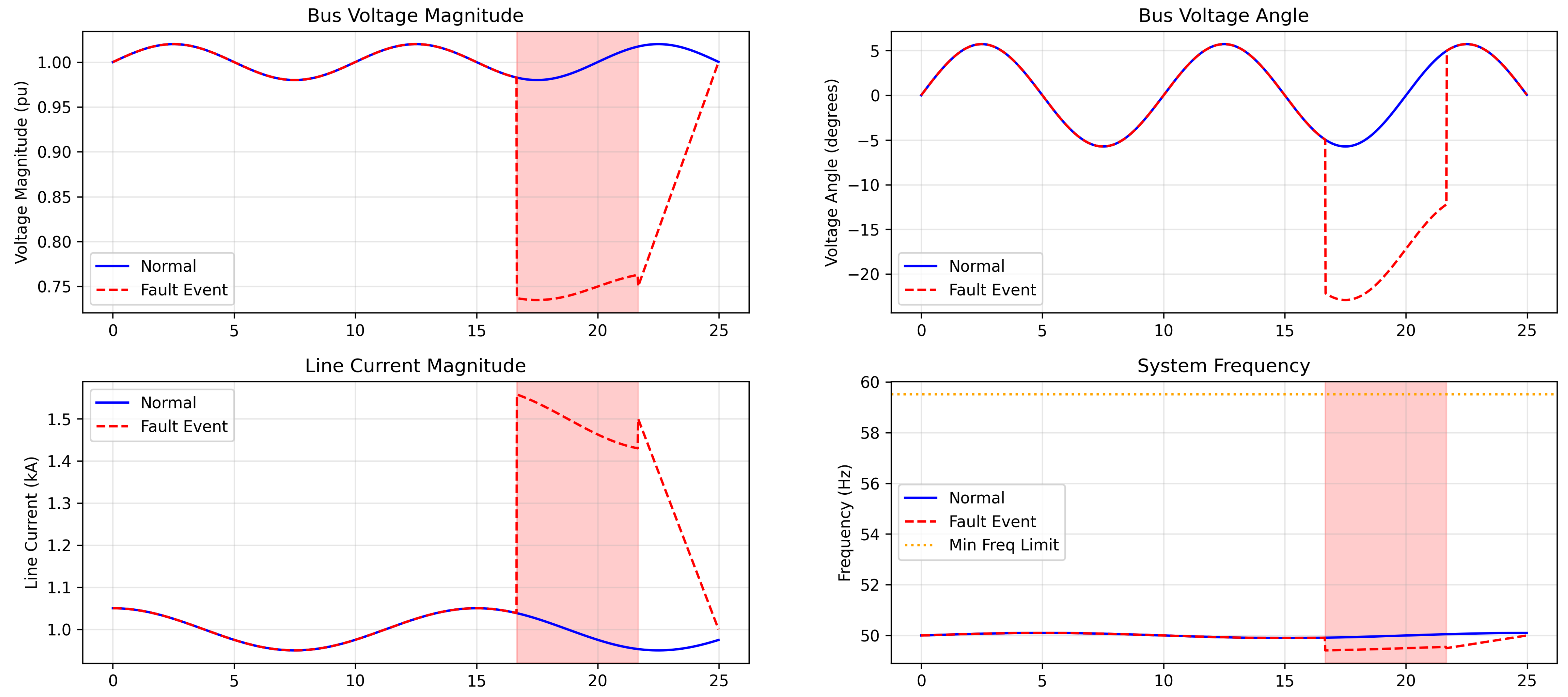
- Test Systems: IEEE `9`, `14`, `39`, `118`, and `300`-bus standard test systems.

Data Source

- Data Source: `OPSD Germany` historical data used for realistic renewable and EV load patterns.

Input: High-Frequency Event Signatures

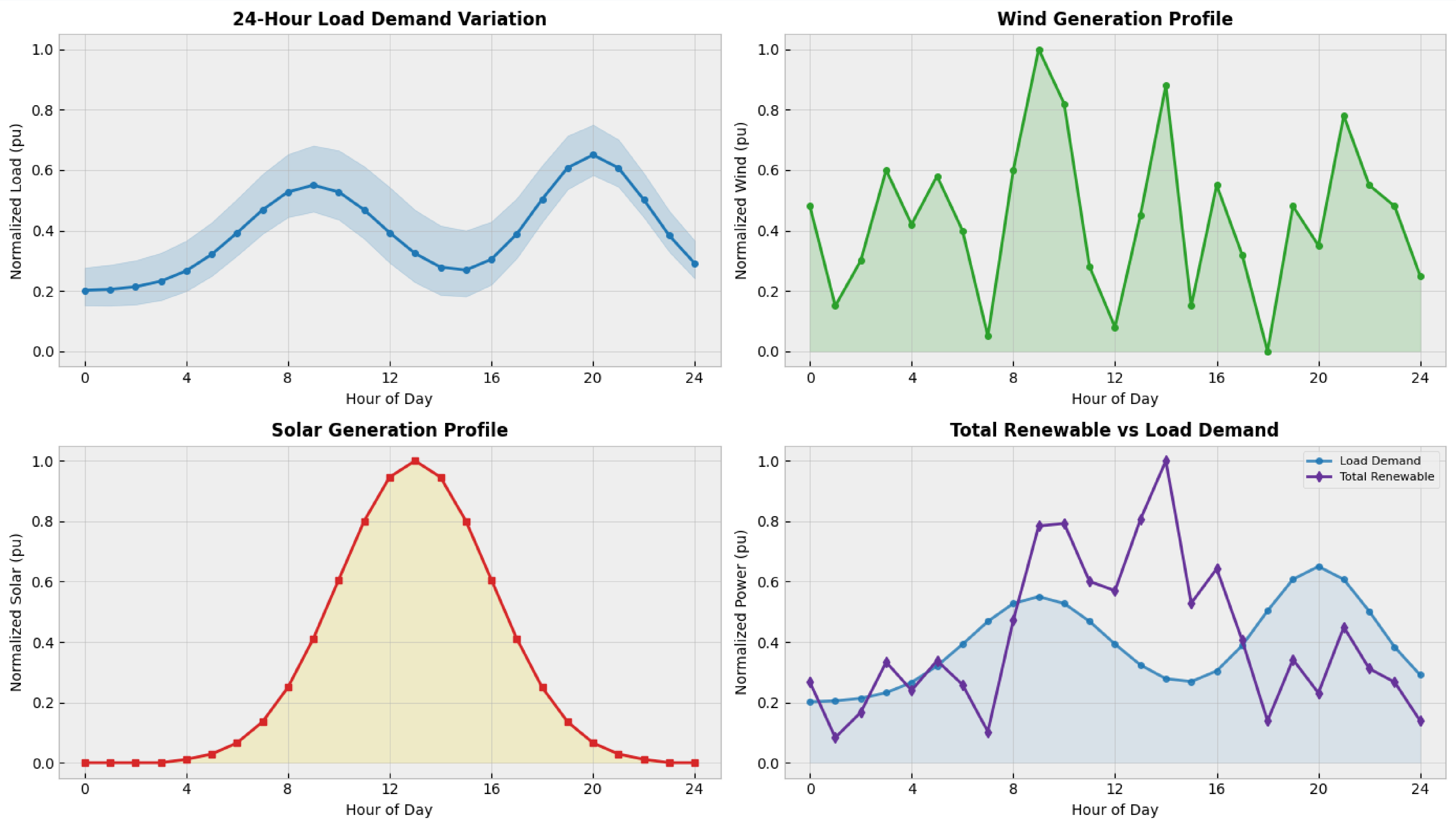
Raw PMU data capturing millisecond-scale fault events for disturbance detection.



- Raw PMU input for CNN–LSTM disturbance detection.

Input: 24-Hour Dynamic Disturbances

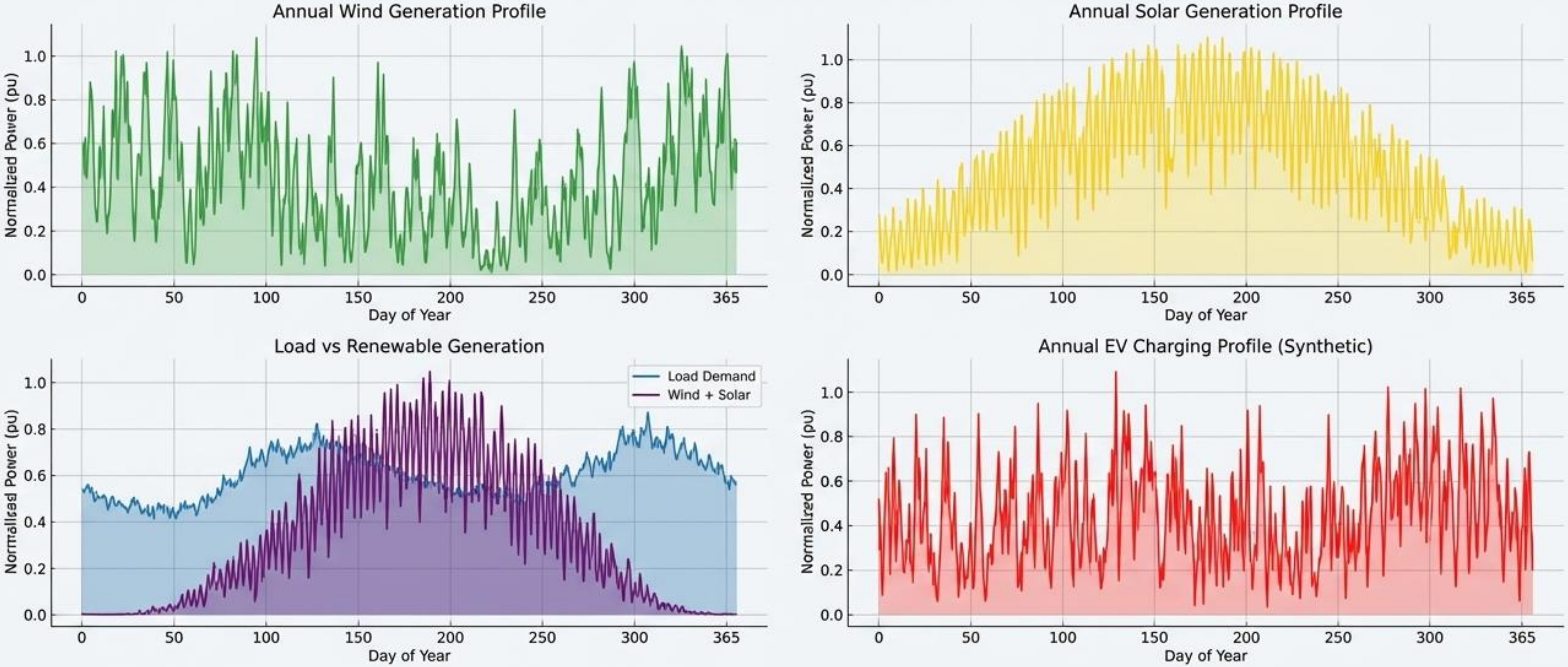
Realistic daily fluctuations in load and renewable generation create a volatile environment.



- Disturbance forcing functions for the Reinforcement Learning environment.

Input: Annual Renewable & Load Profiles

Year-long datasets designed for comprehensive stress-testing against seasonal variations.



- Long-term variability dataset for stress-testing.

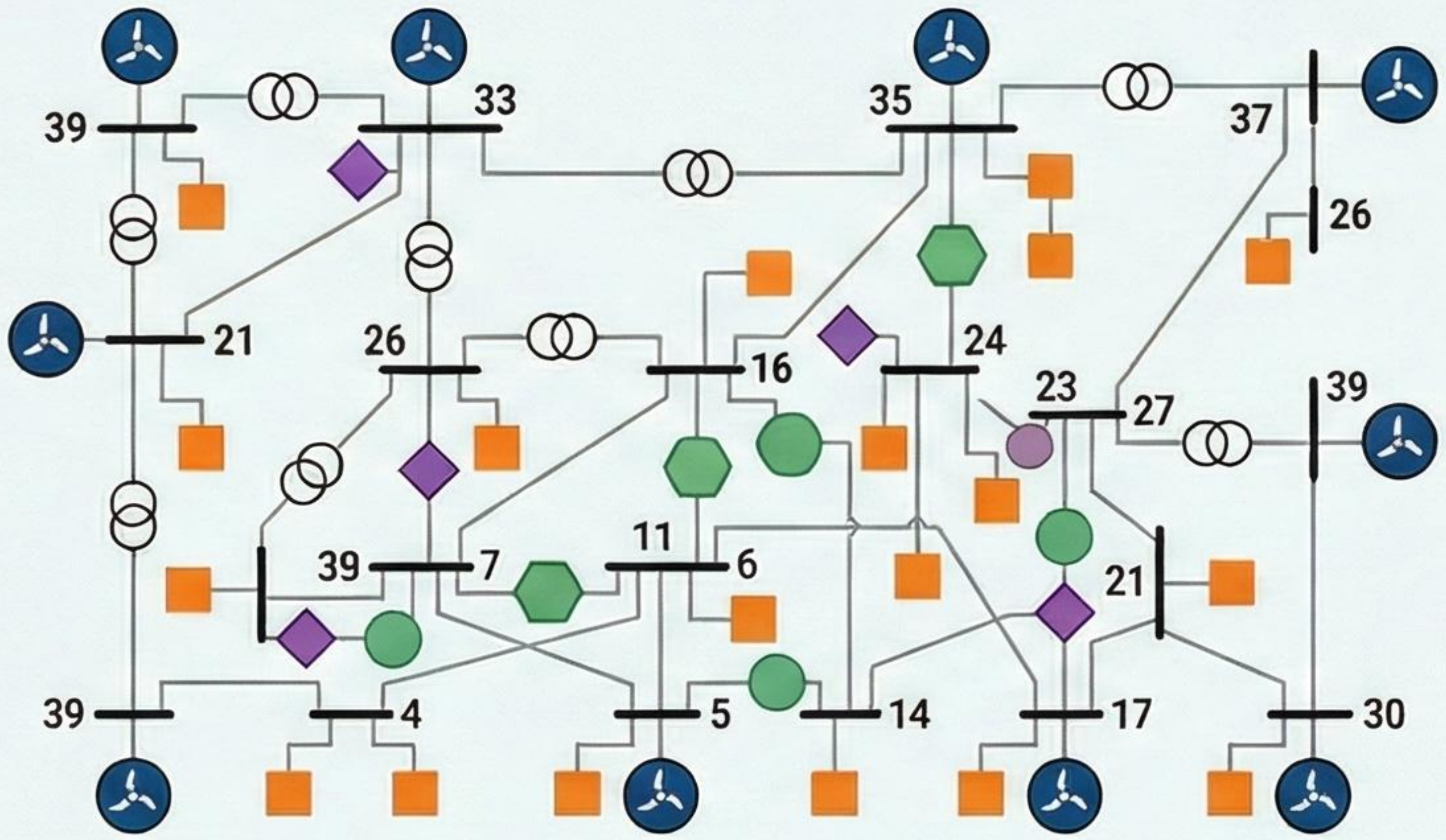
The IEEE 39-Bus System: A Cognitive Control Testbed

AI Controllers (CAPSM)

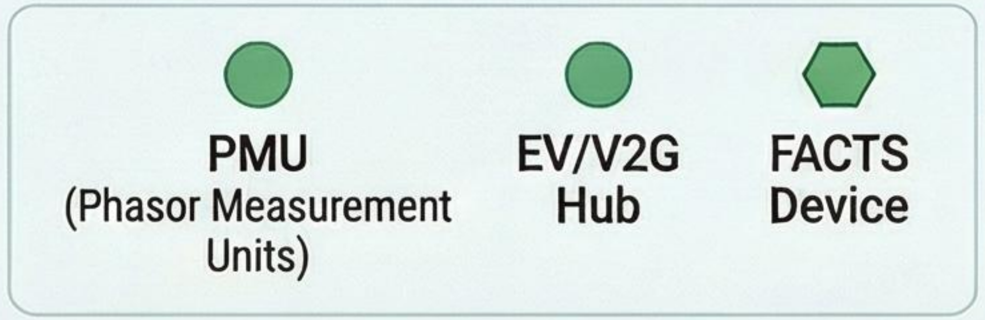
- **System-1: Reflexive Control** (Fast CNN-LSTM, millisecond response)
- **System-2: Planning Control** (Quantum-Inspired RL, long-term scheduling)
- **Metacognitive Arbitrator** (Intelligent choice between System-1 & 2 based on confidence)

Testbed Overview

- **IEEE 39-Bus New England System** (Standard research model)
- **39 Buses, 10 Generators, 21 Aggregated Loads, 46 Transmission Lines, 12 Transformers**
- Primary benchmark for all CAPSM experiments



Schematic topology of **IEEE 39-bus New England transmission system**



Data Inputs

- **Renewable & Load Profiles** (OPSD-derived patterns)
- **EV Fleet Simulation** (Synthetic charging profiles)
- **Grid Measurement Data** (High-rate PMU & SCADA streams)

Performance Metrics

- **Fault Detection Performance** (Accuracy & detection time in milliseconds)
- **Safety Constraint Adherence** (Violation rates for voltage, loading, frequency, EV SoC)
- **Cyber-Physical Stability** (Resilience under network latency, FDI, DoS attacks)

Safety & Operational Limits

- **Voltage & Thermal Limits** (0.95–1.05 pu, 100% line loading)
- **Frequency & Ramping Limits** (59.5–60.5 Hz, generator ramp rates)
- **EV Battery Constraints** (SoC bounded for health)

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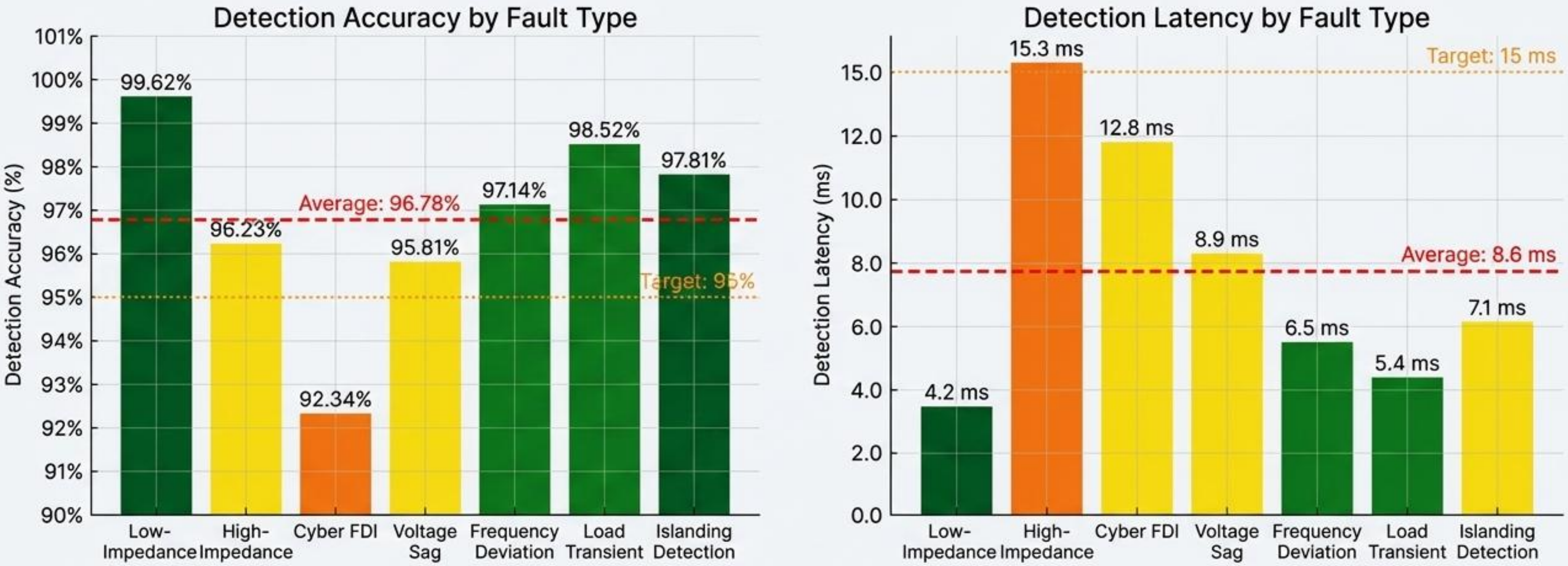
System-1: High-Speed Fault Detection & Classification

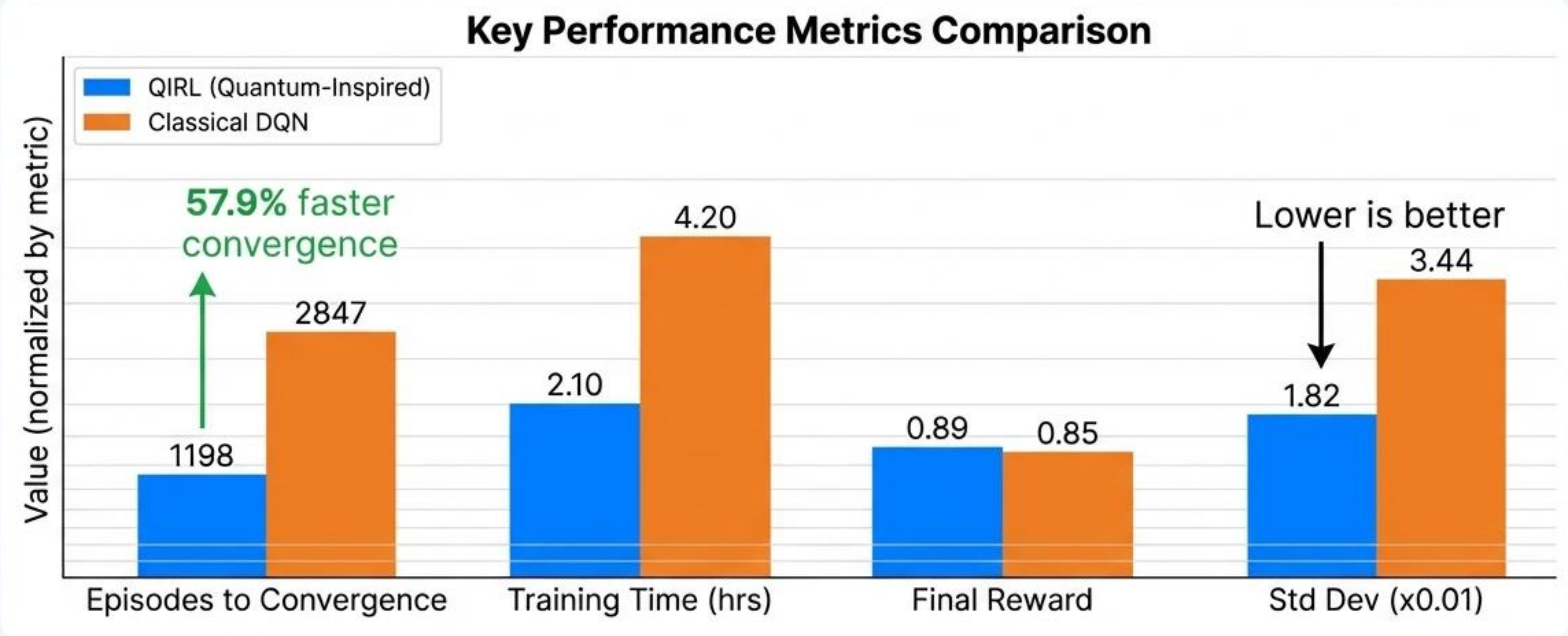
Achieving exceptional accuracy and latency, far exceeding industry requirements.



- Average detection latency of 8.6 ms with 96.78% accuracy across 7 fault types.

System-2: Quantum-Inspired RL Performance

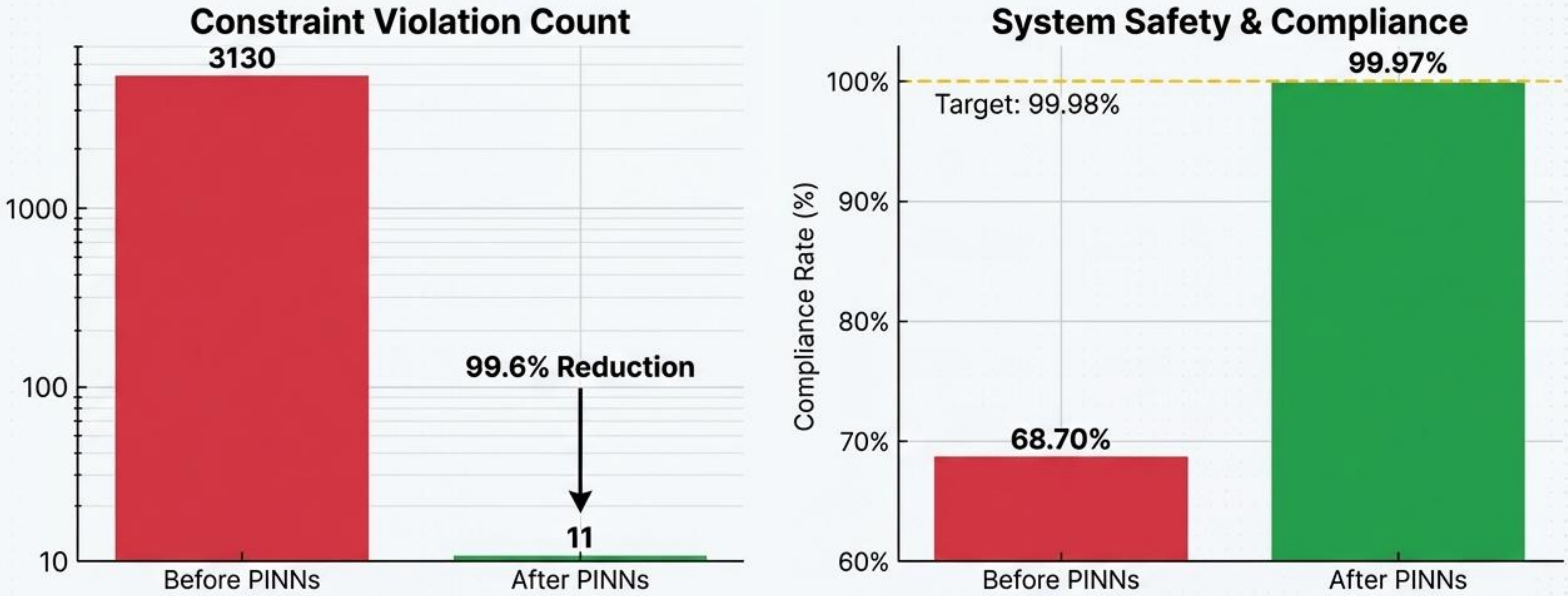
QIRL agent converges 57.9% faster and achieves a more stable control policy than classical DQN.



- QIRL converges faster and more stably than classical DQN.

System-3: Physics-Informed Constraint Adherence

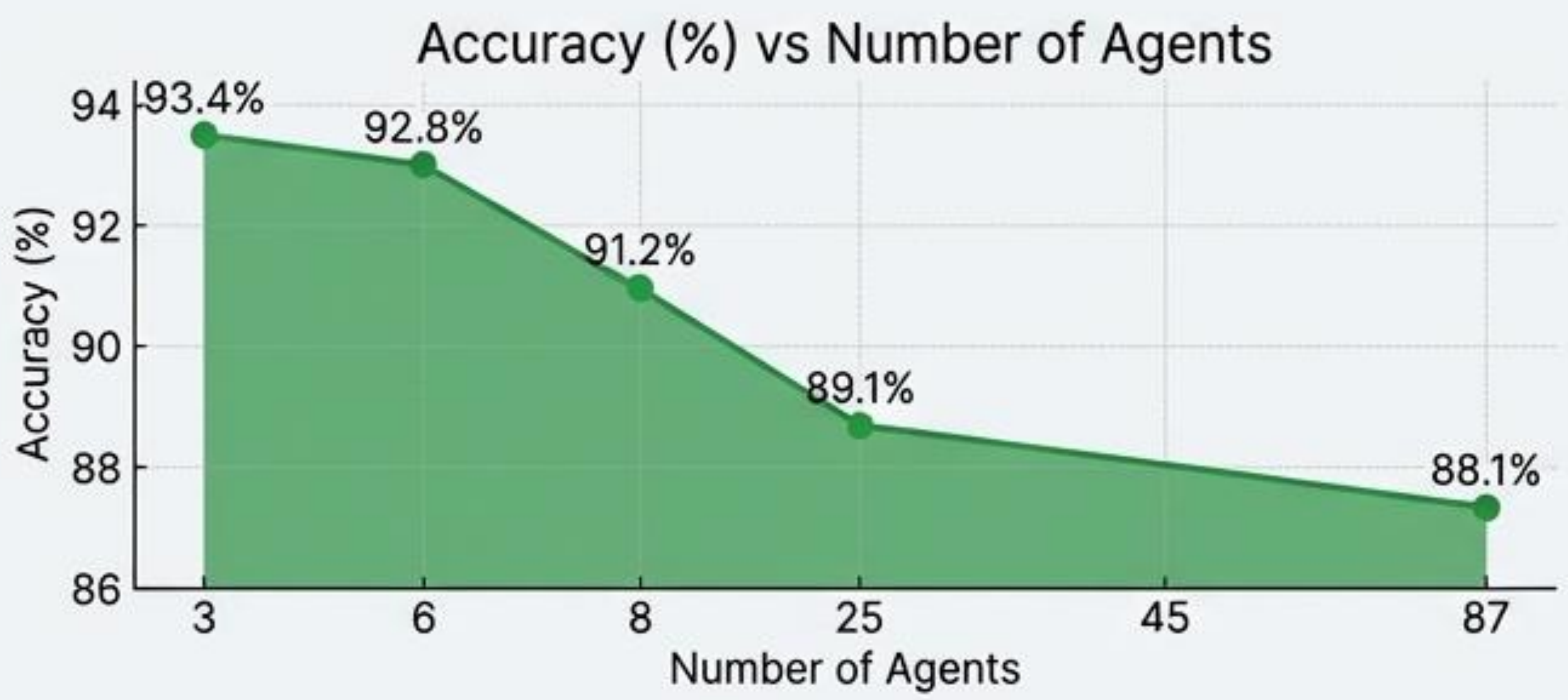
PINNs reduce operational constraint violations by over 99.6%, ensuring system safety.

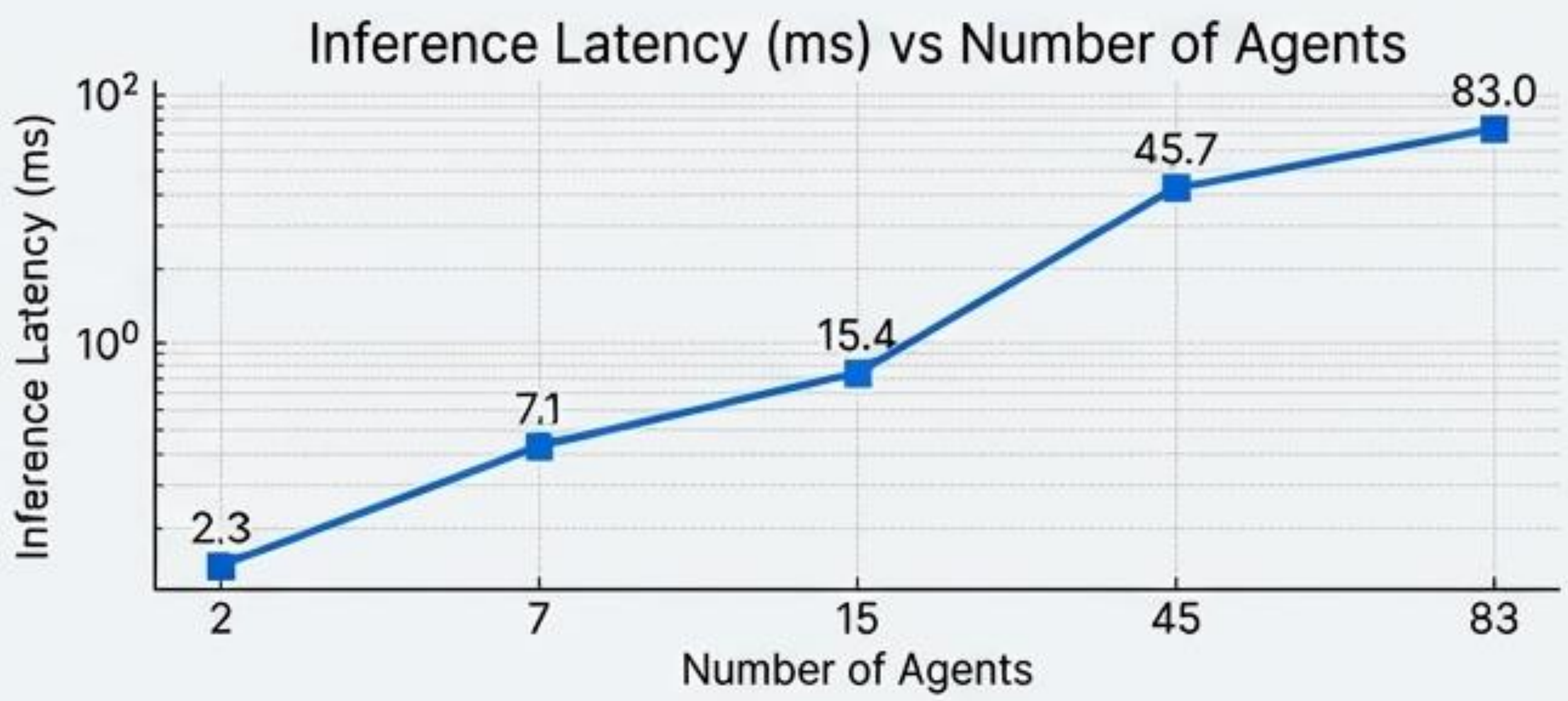


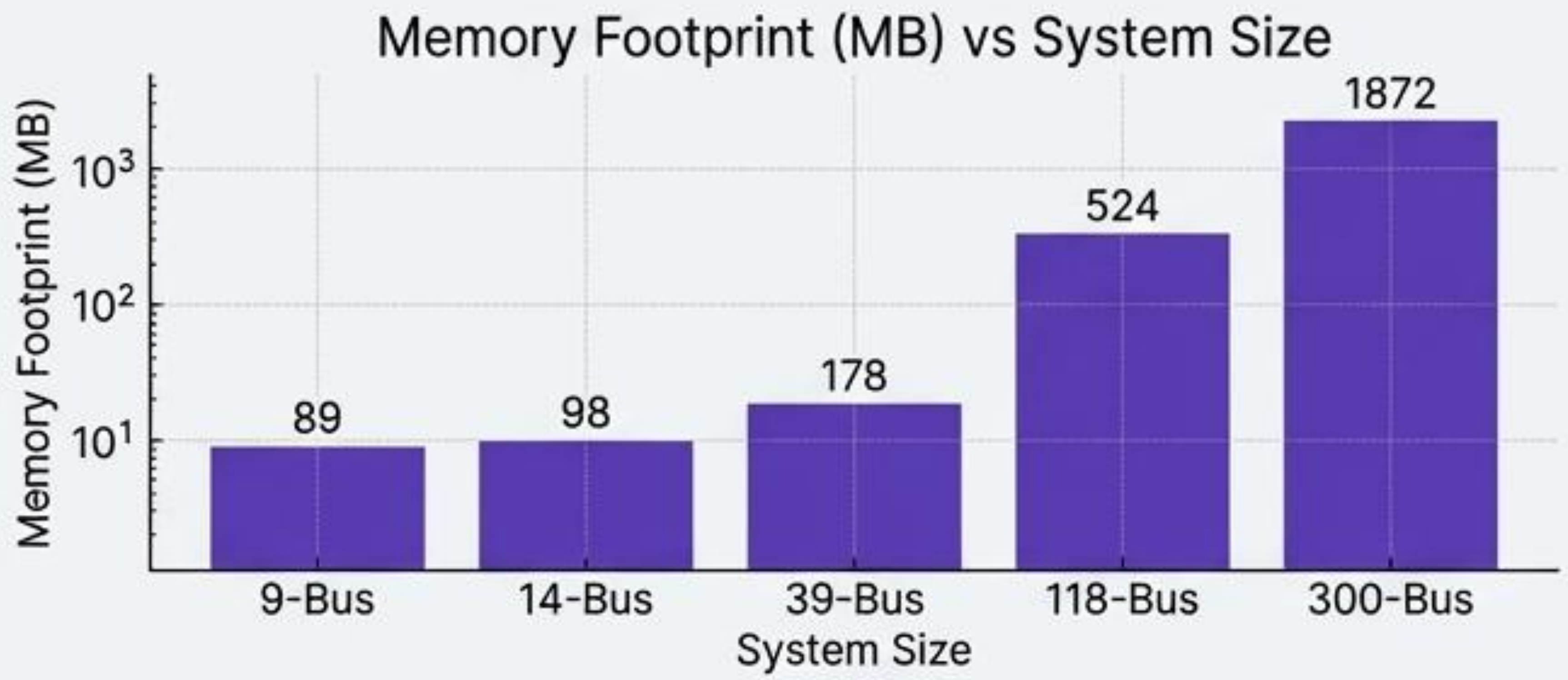
- PINNs virtually eliminate constraint violations, pushing compliance to 99.97%.

System-4: Hierarchical Multi-Agent RL Scalability

Performance scales effectively from small test systems (9-Bus) to large, realistic grids (300-Bus).







HMARL SCALABILITY SUMMARY

IEEE 39-Bus: 8 Agents, 91.23% Accuracy, 15.4 ms Inference

Key scaling characteristics:
Accuracy degradation: ~0.05% per agent
Inference time: ~O(n^1.8) scaling
Memory usage: ~O(n^2) worst-case
Practical limit: ~200-300 buses

- Maintains high accuracy and manageable latency while **scaling** from 9-bus to 300-bus systems.

CAPSM: A Brain-Inspired Framework for the Future Power Grid

How CAPSM mimics the human brain's two-system thinking to solve the modern power grid's core challenges.

The Modern Grid's Dilemma

50%+
Renewables by 2035

Increasing reliance on intermittent sources introduces extreme variability and uncertainty.

300+
Million EVs by 2030

Mass-scale electric vehicle charging adds complex, dynamic demand patterns.

Core Conflict: Speed vs. Optimality

The grid must balance ultra-fast fault prevention with slower global optimization.

Metacognitive Arbitration: Executive Function

Supervices the two systems, ensuring the right response at the right time.

System-1: Reflexive Layer (< 5 ms)

Provides fast, automatic, and intuitive responses to emergencies like sudden faults.

System-2: Deliberative Layer (< 50 ms)

Handles slower, analytical planning for economic dispatch and demand optimization.

Technological Pillars & Performance

Advanced AI for Safe, Stable Control

Uses Quantum-inspired RL, Physics-Informed Neural Networks, and built-in cyber resilience.

< 0.2%
Constraint Violation Rate

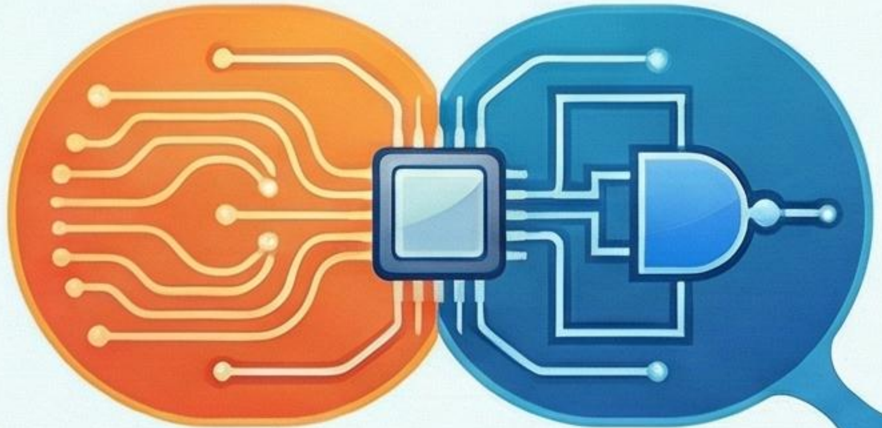
Demonstrates extremely high levels of safe and physically valid grid operation.

Superior Performance

Outperforms prior approaches in speed, safety, and integrated management of resources.

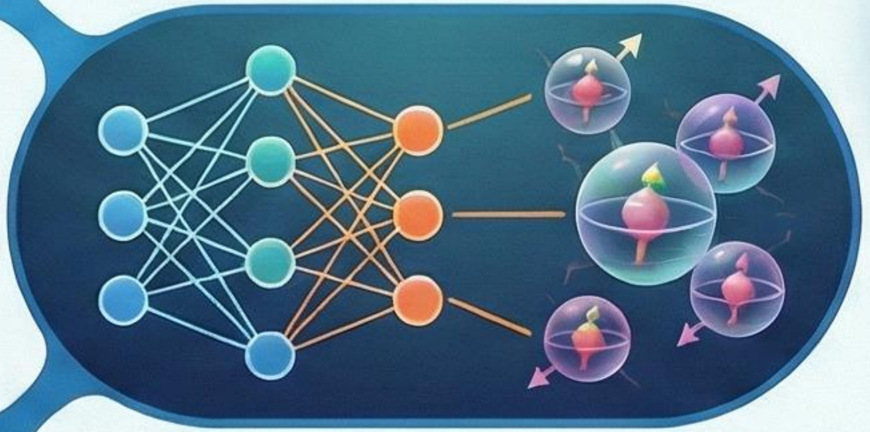
Defining the Boundaries: Scope of the CAPSM PhD Proposal

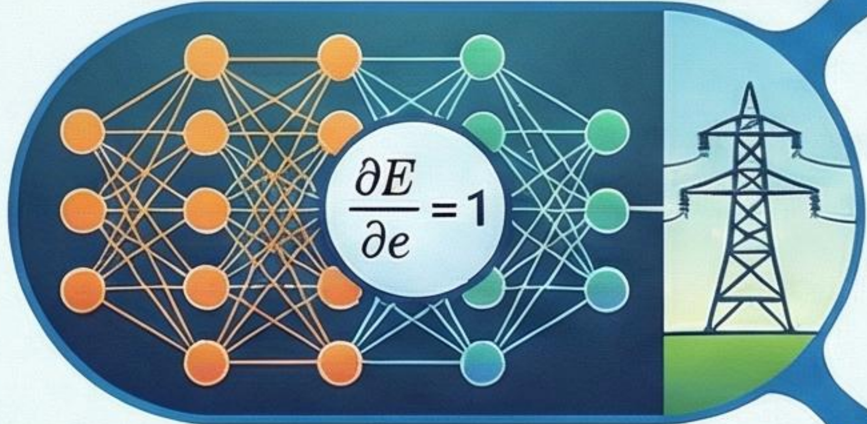
IN SCOPE: Specific Research Areas & Methodologies



Dual-Process Control Architecture

Quantum-Inspired Reinforcement Learning





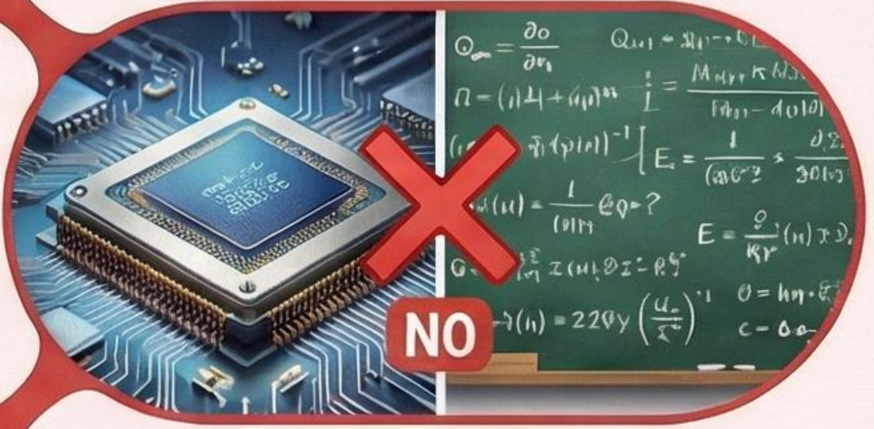
Physics-Informed Neural Networks (PINNs)

Hierarchical Multi-Agent RL (HMARL)

OUT OF SCOPE: Exclusions & Limitations

No Deep Neurocognitive Modelling

No Quantum Hardware or Full Quantum Theory



No National-Scale Market or Regulatory Analysis

No Physical Prototyping or Full Industrial Deployment

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References – Part 1

Publication	Relevancy to Thesis
A proposed controller for real-time management of electrical vehicle battery fleet with MATLAB/SIMULINK *Elsevier, 2025*	Supplies methods for EV/V2G operational control; **forms the basis for EV coordination and System-2 optimization**.
A comprehensive review of the current status of smart grid technologies for renewable energies integration and future trends: The role of machine learning and energy storage, *Energies, 2024*	Provides foundational context on smart grid challenges; **key research gaps motivating the CAPSM architecture are identified**.
Enhancing fire protection in electric vehicle batteries through thermal energy storage and machine learning using Simulink, Fire, 2024	Contributes safety-critical modeling; **supports PINN-based constraint enforcement and safety layers**.
Evaluating solar power forecasting robustness: A comparative analysis of XGBoost, RNN, KNN, RF, and LSTM with emphasis on lagged steps sensitivity and cross-validation techniques, *IEEE Conference, 2024*	Offers renewable variability insights; **provides disturbance inputs for RL and planning layers**.
Enhancing wind power forecasting accuracy in Canada using a solar data-enhanced hybrid machine learning model: Integrating ANN, LSTM, and SVR, *IEEE Conference 2024*	Provides enhanced forecasting methods; **strengthens forecast-driven adaptive control**.
Comparative analysis of machine learning techniques for an hour-ahead forecasting of electric vehicle states, *Master Thesis, 2023*	Improves understanding of EV SoC and charging patterns; **supports multi-agent EV modeling and SoC constraints**.
AI-driven control strategies for FACTS devices in power quality management: A comprehensive review, *Applied Science, 2025*	Identifies FACTS control limitations; **directly motivates hierarchical multi-agent coordination (HMARL)**.
Climate-adaptive residential demand response with power quality-aware optimization using distributed generation systems, *Electronics, 2025*	Introduces multi-objective optimization; **supports System-2 cost–PQ–stability trade-off modeling**.

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References – Part 2

Prior Approach	Full Supporting Publications
Centralized MPC / OPF for Power Electronics and Microgrids	**[15]** Model Predictive Control of Power Electronic Systems: Methods, Results, and Challenges (Karamanakos et al.), **[16]** Model Predictive Control Strategies in Microgrids: A Concise Revisit (Shahzad et al.)
Model-Based Deep Reinforcement Learning for Voltage/Power Control	**[17]** Efficient Learning of Voltage Control Strategies via Model-based Deep Reinforcement Learning (Hossain et al.)
Safe RL with Barrier Certificates / Lyapunov Stability	**[18]** A Barrier-Certificated Reinforcement Learning Approach for Enhancing Power System Transient Stability (Zhao et al.), **[19]** Safe Reinforcement Learning for Power System Control: A Review (Yu et al.)
Physics-Informed Neural Networks (PINNs)	**[20]** Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations (Raissi et al.)
Heuristic / Classical Frequency and Protection Control	**[21]** Robust Power System Frequency Control (Bevrani)
Multi-Agent RL for Energy Networks and Smart Communities	**[22]** Multi-Agent Reinforcement Learning for Energy Networks: Computational Challenges, Progress and Open Problems (Keren et al.), **[23]** Multi-Agent Reinforcement Learning for Smart Community Energy Management (Wilk et al.)
Cybersecurity and Smart Grid Anomaly Detection	**[24]** A Comprehensive Review of Cyber-Attacks and Defense Mechanisms for Improving Security in Smart Grid Energy Systems: Past, Present and Future (Ghiasi et al.), **[25]** Security Risk Models Against Attacks in Smart Grid Using Big Data and Artificial Intelligence (Ghadi et al.)
EV / V2G Dispatch and Charging MPC	**[26]** Model Predictive Control of Vehicle Charging Stations in Grid-Connected Microgrids: An Implementation Study (Hermans

The Defensible PhD: A 24-Month Strategic Roadmap

Months 1-4: Foundation & Framing

Anchor in Power Systems, Not AI
Start by defining real-world power quality challenges and control limitations.

Formalize the Research Gap
Explicitly identify what existing controllers **cannot do** to justify the research.

Existing Controllers
Unmet Needs

Make It Scientific, Not Just Engineering
Formulate testable hypotheses and define clear, quantitative evaluation metrics.

Months 5-14: Core Technical Development

Control Framework
Hierarchical Structure
Separation of Control vs. Learning

Define the Control Architecture
Detail the proposed framework, its hierarchy, and the separation of control vs. learning.

$\Sigma \int f(x)$

Power System Model & Control Equations

Build the Mathematical Backbone
This is the most critical phase; model the power system and control equations.

Augment Control With AI

Physics-informed Safety Bounds
AI Augmentation
Augmented Control Output

Defensibly claim AI *augments* control, implementing it within physics-informed safety bounds.

Months 14-24: Validation & Finalization

Prove Robustness Through Benchmarking
Conduct comparative studies and sensitivity analysis across standard test systems.

Increase Credibility by Admitting Limits
Discuss failure cases, generalizations, and practical limitations of the approach.

Consolidate & Disseminate
Focus on final dissertation writing, journal publications, and defense preparation.